

Day 4 Tasks

Exercise 1:

Program :

```
1  list1 = [3, 6, 9, 12, 15, 18, 21]
2  list2 = [4, 8, 12, 16, 20, 24, 28]
3
4  odd_elements = list1[1::2]
5  print("Element at odd-index positions from list one")
6  print(odd_elements)
7
8  even_elements = list2[0::2]
9  print("Element at even-index positions from list two")
10 print(even_elements)
11
12 res = []
13 res.extend(odd_elements)
14 res.extend(even_elements)
15
16 print("Printing Final third list")
17 print(res)
```

Output:

```
Element at odd-index positions from list one
[6, 12, 18]
Element at even-index positions from list two
[4, 12, 20, 28]
Printing Final third list
[6, 12, 18, 4, 12, 20, 28]
PS C:\Users\GANESH\OneDrive\Desktop\DS Exercises\Day 3>
```

Exercise 2:

Program:

```
1 sample_list = [34, 54, 67, 89, 11, 43, 94]
2
3 print("Original list:", sample_list)
4
5 element = sample_list.pop(4)
6 print("List After removing element at index 4:", sample_list)
7
8 sample_list.insert(2, element)
9 print("List after Adding element at index 2:", sample_list)
10
11 sample_list.append(element)
12 print("List after Adding element at last:", sample_list)
```

Output:

```
Original list: [34, 54, 67, 89, 11, 43, 94]
List After removing element at index 4: [34, 54, 67, 89, 43, 94]
List after Adding element at index 2: [34, 54, 11, 67, 89, 43, 94]
List after Adding element at last: [34, 54, 11, 67, 89, 43, 94, 11]
PS C:\Users\GANESH\OneDrive\Desktop\DS Exercises\Day 3>
```

Exercise 3:

Program:

```
1 sample_list = [11, 45, 8, 23, 14, 12, 78, 45, 89]
2 print("Original list:", sample_list)
3
4 length = len(sample_list)
5 chunk_size = length // 3
6
7 start = 0
8 end = chunk_size
9
10 for i in range(3):
11     chunk = sample_list[start:end]
12     print(f"Chunk {i+1}:", chunk)
13     print("After reversing it:", list(reversed(chunk)))
14
15     start = end
16     end += chunk_size
```

Output:

```
Original list: [11, 45, 8, 23, 14, 12, 78, 45, 89]
Chunk 1: [11, 45, 8]
After reversing it: [8, 45, 11]
Chunk 2: [23, 14, 12]
After reversing it: [12, 14, 23]
Chunk 3: [78, 45, 89]
After reversing it: [89, 45, 78]
PS C:\Users\GANESH\OneDrive\Desktop\DS Exercises\Day 3>
```

Exercise 4:

Program :

```
1  sample_list = [11, 45, 8, 11, 23, 45, 23, 45, 89]
2  print("Original list:", sample_list)
3
4  count_dict = {}
5
6  for item in sample_list:
7      if item in count_dict:
8          count_dict[item] += 1
9      else:
10         count_dict[item] = 1
11
12  print("Printing count of each item:", count_dict)
```

Output:

```
Original list: [11, 45, 8, 11, 23, 45, 23, 45, 89]
Printing count of each item: {11: 2, 45: 3, 8: 1, 23: 2, 89: 1}
PS C:\Users\GANESH\OneDrive\Desktop\DS Exercises\Day 3>
```

Exercise 5:

Program:

```
1  first_list = [2, 3, 4, 5, 6, 7, 8]
2  second_list = [4, 9, 16, 25, 36, 49, 64]
3
4  result = set(zip(first_list, second_list))
5  print("Result is:", result)
```

Output:

```
Result is: {(7, 49), (2, 4), (4, 16), (8, 64), (6, 36), (3, 9), (5, 25)}  
PS C:\Users\GANESH\OneDrive\Desktop\DS Exercises\Day 3>
```

Exercise 6:

Program:

```
1  first_set = {23, 42, 65, 57, 78, 83, 29}  
2  second_set = {57, 83, 29, 67, 73, 43, 48}  
3  
4  intersection = first_set.intersection(second_set)  
5  print("Intersection is:", intersection)  
6  
7  for item in intersection:  
8      first_set.remove(item)  
9  
10 print("First Set after removing common element:", first_set)
```

Output:

```
Intersection is: {57, 83, 29}  
First Set after removing common element: {65, 23, 42, 78}  
PS C:\Users\GANESH\OneDrive\Desktop\DS Exercises\Day 3>
```

Exercise 7:

Program:

```
1  first_set = {57, 83, 29}  
2  second_set = {57, 83, 29, 67, 73, 43, 48}  
3  
4  print("First set is subset of second set -", first_set.issubset(second_set))  
5  print("Second set is subset of first set -", second_set.issubset(first_set))  
6  
7  print("First set is superset of second set -", first_set.issuperset(second_set))  
8  print("Second set is superset of first set -", second_set.issuperset(first_set))  
9  
10 if first_set.issubset(second_set):  
11     first_set.clear()  
12  
13 if second_set.issubset(first_set):  
14     second_set.clear()  
15  
16 print("First Set:", first_set)  
17 print("Second Set:", second_set)
```

Output:

```
First set is subset of second set - True
Second set is subset of first set - False
First set is superset of second set - False
Second set is superset of first set - True
First Set: set()
Second Set: {48, 67, 83, 73, 43, 29, 57}
PS C:\Users\GANESH\OneDrive\Desktop\DS Exercises\Day 3>
```

Exercise 8:

Program:

```
1 roll_number = [47, 64, 69, 37, 76, 83, 95, 97]
2 sample_dict = {'Jhon': 47, 'Emma': 69, 'Kelly': 76, 'Jason': 97}
3
4 roll_number[:] = [item for item in roll_number if item in sample_dict.values()]
5
6 print("After removing unwanted elements from list:", roll_number)
```

Output:

```
After removing unwanted elements from list: [47, 69, 76, 97]
PS C:\Users\GANESH\OneDrive\Desktop\DS Exercises\Day 3>
```

Exercise 9:

Program:

```
1 v speed = {
2     'jan': 47, 'feb': 52, 'march': 47, 'April': 44,
3     'May': 52, 'June': 53, 'july': 54, 'Aug': 44, 'Sept': 54
4 }
5
6 unique_values = []
7
8 v for val in speed.values():
9     v if val not in unique_values:
10         unique_values.append(val)
11
12 print("Unique list:", unique_values)
```

Output:

```
Unique list: [47, 52, 44, 53, 54]  
PS C:\Users\GANESH\OneDrive\Desktop\DS Exercises\Day 3>
```

Exercise 10:

Program:

```
1  sample_list = [87, 45, 41, 65, 94, 41, 99, 94]  
2  
3  unique_list = list(set(sample_list))  
4  print("Unique items:", unique_list)  
5  
6  t = tuple(unique_list)  
7  print("Tuple:", t)  
8  
9  print("Min:", min(t))  
10 print("Max:", max(t))
```

Output:

```
Unique items: [65, 99, 41, 45, 87, 94]  
Tuple: (65, 99, 41, 45, 87, 94)  
Min: 41  
Max: 99  
PS C:\Users\GANESH\OneDrive\Desktop\DS Exercises\Day 3>
```